Engineering Portfolio — Prerana Kapadnis

Full-Stack Engineer & AI Builder | MERN, PERN, & AI/ML

Formally designing high-throughput data pipelines and fine-tuning LLMs. Informally here to experiment, try things that shouldn't work, and see what happens when you push tech to its limits.

📫 **Connect:** [LinkedIn] | [GitHub] | [Email]

🎯 Overview

This notebook is a living, executable showcase of my core engineering milestones, architecture choices, and experimental projects. **Run the cells below** to see live pipeline simulations, optimization metrics, and model evaluations.

[Initialization] Run this cell to boot up Prerana

[6]
✓ 0s

```python
# @title [Initialization] Run this cell to bo
import time
import sys
import pandas as pd
import numpy as np
import plotly.graph_objects as go
import plotly.express as px
import ipywidgets as widgets
from IPython.display import display, HTML, c

print("🚀 Initializing Portfolio Core Engine.
time.sleep(0.5)
print(f"🖥️ System Python Version: {sys.versio
print("📦 Core Modules Loaded: Pandas, Plotly
print("✅ Status: Ready to Explore.")
```

```
🚀 Initializing Portfolio Core Engine...
🖥️ System Python Version: 3.12.13
📦 Core Modules Loaded: Pandas, Plotly, IPyWi
✅ Status: Ready to Explore.
```

[Initialization] Run this cell to boot up Prerana's Engineering Portfolio Dashboard

[6]
✓ 0s

```python
# @title [Initialization] Run this cell to b
import time
import sys
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🔍 Interactive Project Filter

```python
# @title 🔍 Interactive Project Filter { display-mode: "form" }

# Data array representing your deep-dive execution history
projects_db = [
    {"name": "Genesis (Autonomize.AI)", "domain": "Fullstack / Data Pipelines", "tech": "PERN, FastAPI, Apache Kafka", "impact": "Ingested streaming data with real-time JSON parsing and built a robust document validation pipeline resolving critical webhooks."},
    {"name": "Capgemini UI Frameworks", "domain": "Fullstack / Data Pipelines", "tech": "JavaScript,
```

```
HTML/CSS, Docker", "impact": "Enhanced internal web
app UI, supported Agile sprints via feature dev, and
implemented JUnit testing."},
    {"name": "Bash Scripting Assistant", "domain":
"AI / ML", "tech": "QLoRA, CodeLlama-7B, Phi-3,
Streamlit", "impact": "Fine-tuned CodeLlama-7B on
custom NL-to-bash dataset. Achieved 0.9 CodeBLEU and
0.8 ROUGE alignment."},
    {"name": "AI-Driven B2B Pharmacy System",
"domain": "AI / ML", "tech": "Python, Flask,
PostgreSQL", "impact": "Parallelized model training
achieving a 30% reduction in runtime. Published
research paper."},
    {"name": "Network Traffic Analyzer", "domain":
"Low-Level & Logic", "tech": "C++, Wireshark, Socket
Programming", "impact": "Built a multithreaded
network packet analysis tool capturing, filtering,
and diagnosing TCP/UDP traffic."},
    {"name": "Automated Grading System", "domain":
"Backend Engine", "tech": "Java, Spring Boot,
Hibernate, MySQL", "impact": "Engineered scalable
backend achieving over 95% test coverage using JUnit/
Mockito. Reduced query response times by 40%."}
]

dropdown = widgets.Dropdown(
    options=['All Domains', 'Fullstack / Data
Pipelines', 'AI / ML', 'Low-Level & Logic', 'Backend
Engine'],
    value='All Domains',
    description='Domain:',
    disabled=False,
)

def display_filtered_projects(change):
    clear_output(wait=True)
    display(dropdown)
    selected = change['new'] if change else 'All
Domains'

    print(f"\n--- Showing Projects for: {selected}
```

```
---\n")
    for p in projects_db:
        if selected == 'All Domains' or p['domain']
== selected:
            html_box = f"""
            <div style="border-left: 4px solid
#1a73e8; padding: 10px; margin-bottom: 15px;
background-color: #f8f9fa; border-radius: 0 4px 4px
0;">
                <b style="font-size: 16px; color:
#202124;">{p['name']}</b> <span style="color:
#5f6368; font-size: 12px;">({p['tech']})</span>
                <p style="margin: 5px 0 0 0; color:
#3c4043; font-size: 14px;"><b>Impact Execution:</b>
{p['impact']}</p>
            </div>
            """
            display(HTML(html_box))

dropdown.observe(display_filtered_projects,
names='value')
display(dropdown)
display_filtered_projects(None)
```

Domain: All Domains

```
--- Showing Projects for: All Domains ---
```

Genesis (Autonomize.AI) (PERN, FastAPI, Apache Kafka)
Impact Execution: Ingested streaming data with real-time JSON p

Capgemini UI Frameworks (JavaScript, HTML/CSS, Docker)
Impact Execution: Enhanced internal web app UI, supported Agile

Bash Scripting Assistant (QLoRA, CodeLlama-7B, Phi-3, Streamlit)
Impact Execution: Fine-tuned CodeLlama-7B on custom NL-to-ba

AI-Driven B2B Pharmacy System (Python, Flask, PostgreSQL)
Impact Execution: Parallelized model training achieving a 30% re

Network Traffic Analyzer (C++, Wireshark, Socket Programming)
Impact Execution: Built a multithreaded network packet analysis t

Automated Grading System (Java, Spring Boot, Hibernate, MySQL
Impact Execution: Engineered scalable backend achieving over 9

Domain: Fullstack / Data Pipelines

```
--- Showing Projects for: Fullstack / Data Pipeline
```

Genesis (Autonomize.AI) (PERN, FastAPI, Apache Kafka)

Impact Execution: Ingested streaming data with real-time JSON p

Capgemini UI Frameworks (JavaScript, HTML/CSS, Docker)

Impact Execution: Enhanced internal web app UI, supported Agile

Domain: AI / ML

```
--- Showing Projects for: AI / ML ---
```

Bash Scripting Assistant (QLoRA, CodeLlama-7B, Phi-3, Streamlit)

Impact Execution: Fine-tuned CodeLlama-7B on custom NL-to-ba

AI-Driven B2B Pharmacy System (Python, Flask, PostgreSQL)

Impact Execution: Parallelized model training achieving a 30% re

Domain: Low-Level & Logic

```
--- Showing Projects for: Low-Level & Logic ---
```

Network Traffic Analyzer (C++, Wireshark, Socket Programming)

Impact Execution: Built a multithreaded network packet analysis t

Domain: Backend Engine

```
--- Showing Projects for: Backend Engine ---
```

Automated Grading System (Java, Spring Boot, Hibernate, MySQl
Impact Execution: Engineered scalable backend achieving over 9

🧪 Deep Tech & AI/ML Projects

🤖 Bash Scripting Assistant (Fine-Tuning CodeLlama)

- [cite_start]**Core Work:** Fine-tuned CodeLlama-7B on a custom natural language-to-bash dataset using **QLoRA** for efficient memory optimization[cite: 36, 38].
- [cite_start]**Results:** Achieved a **0.9 CodeBLEU** and **0.8 ROUGE** score, proving immense semantic alignment[cite: 39].

💊 AI-Driven B2B Pharmacy Ordering System

- [cite_start]**Core Work:** Designed a modular backend architecture using Flask and PostgreSQL[cite: 33]. [cite_start]Parallelized model training to achieve a **30% reduction in training runtime**[cite: 35].
- [cite_start]**Publication:** Published a formal research paper on this framework[cite: 35].

📊 Optimization & Impact Metrics

[8]
✓ 0s

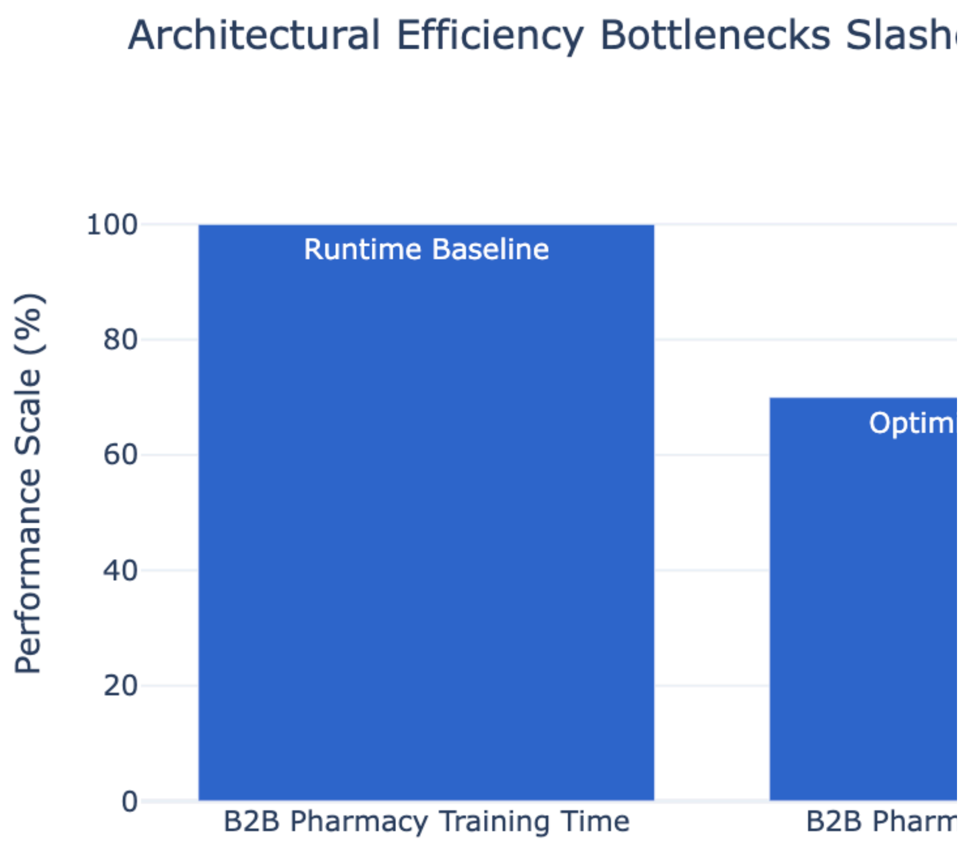
```
# @title 📊 Optimization & Impact Metrics { display

# Data metrics pulled directly from your architectu
metrics_data = {
    'Metric Showcase': [
        'B2B Pharmacy Training Time', 'B2B Pharmacy
        'Grading Sys Response Time', 'Grading Sys (
        'Grading Sys Test Coverage'
    ],
    'Value': [100, 70, 100, 60, 95], # Representing
    'Type': ['Runtime Baseline', 'Optimized (-30%)'
    'Project': ['Pharmacy System', 'Pharmacy System
}

df = pd.DataFrame(metrics_data)

fig = px.bar(
    df,
    x='Metric Showcase',
    y='Value',
    color='Project',
    text='Type',
    title="Architectural Efficiency Bottlenecks Sla
    labels={'Value': 'Performance Scale (%)'},
    color_discrete_sequence=px.colors.qualitative.G
)

fig.update_traces(textposition='inside', textfont_s
fig.update_layout(
    template='plotly_white',
    height=450,
    xaxis_title="",
    yaxis_range=[0, 110]
)
fig.show()
```



🏗️ Core Engineering Mentality: Stream Pipelines